# FINAL PROGRESS REPORT

**Project:** Uncertainty-Aware Contextual Recognition of Sign Language Sequences
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## SUMMARY OF WORK COMPLETED:

Since the first progress report , we have significantly extended our system from isolated gesture recognition to a structured word-level sign recognition pipeline using deep learning and probabilistic sequence modelling.

Initially, we implemented a baseline system for single-letter (ASL alphabet) recognition using a Convolutional Neural Network (CNN) combined with Hidden Markov Model (HMM) decoding. An OpenCV-based live demo was also developed to perform real-time gesture recognition using a webcam. This stage helped us validate the integration of CNN-based visual recognition with probabilistic decoding and understand how uncertainty in predictions can be handled.

Subsequently, we transitioned to a more realistic dataset (WLASL) for word-level recognition. A complete data pipeline was built, including dataset organization, preprocessing, frame extraction, and a custom PyTorch dataloader. A ResNet18-based CNN model was trained to classify sign videos.
A major improvement was achieved by utilizing temporal annotations in the dataset to extract only the relevant sign segments from each video. This reduced noise in the input data and led to a strong baseline performance of **98.06% accuracy**.

To incorporate uncertainty-aware sequence modelling, the trained CNN was used to generate frame-level probability distributions over sign classes. These probabilistic outputs were then used as input to different decoding strategies, including simple temporal aggregation (such as averaging probabilities) and Hidden Markov Model (HMM)-based decoding.

It is important to note that the HMM is not used as a standalone model, but as a probabilistic decoding layer on top of the CNN outputs. For the word-level setting, the HMM was formulated under the assumption that a single sign generates the entire sequence of frames. In this context, the performance of the CNN+HMM system was comparable to that of the CNN baseline model, which had an accuracy of 98.06%, while the performance of other techniques, such as the majority voting method, was slightly less accurate, i.e., 94.17%.

In conclusion, the system combines computer vision and deep learning with probabilistic modeling successfully, thus showing the feasibility of using uncertainty-aware decoding in the context of sign language recognition.

## WORK REMAINING BEFORE FINAL SUBMISSION:

Before the final submission, we plan to:

- Refine the presentation and documentation of the CNN+HMM pipeline.
- Improve alignment with the project proposal by exploring simple sequence-level experiments involving multiple signs.
- Enhance the OpenCV-based demo for clearer demonstration of the final system.
- Add visualizations and structured analysis of model predictions and uncertainty.
- Finalize the report with clear explanations of methodology, results, and limitations.

## CHALLENGES ENCOUNTERED:

The main challenge, however, was how best to process the video data. In the early stages of developing the model, processing the video data led to noisy data, hence affecting the overall

performance of the model. This challenge was addressed by using the temporal annotations to efficiently process the sign segments.

The second challenge was how best to formulate an appropriate HMM. This challenge was addressed by using a naive transition-based formulation of the HMM, which led to poor performance of the model due to noisy predictions. This challenge was addressed by developing an appropriate formulation of the HMM for the isolated word case.Another challenge was processing the intermediate results, e.g., the predictions for the video frames.

**OVERALL PROJECT STATUS:**
At this point, the essential parts of the project have been successfully implemented. The system comprises the following:

- CNN-based visual recognition
- Frame probabilistic predictions
- HMM-based probabilistic decoding
- OpenCV-based real-time demo
- Complete end-to-end pipeline

The remaining work entails refinement and better alignment with the original proposal. The project is on track for a successful final submission.